PAN-CANCER SURVIVAL GENOMICS · DISCOVERY STAGE

Prognostic biomarkers are cancer-type specific

Nearly half of recurrent prognostic genes reverse direction between cancers. Knowing the sign per indication is the asset.

34
TCGA cohorts screened

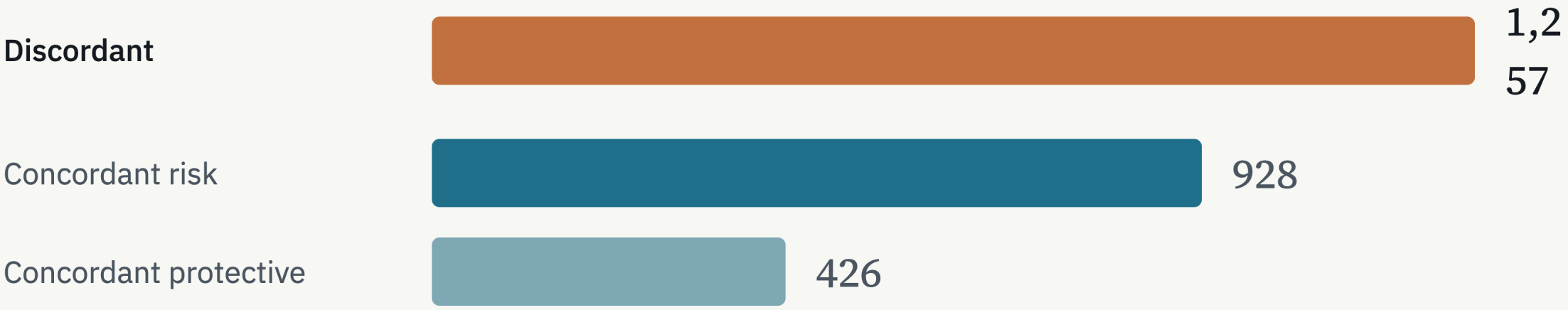
17
Replicated on held-out patients

48.1%
Of recurrent genes reverse direction

Denver Ncube · October 2026 · TCGA RNA and RPPA

Half of recurrent genes reverse direction between cancers

Of 2,611 genes significant in two or more cohorts, 1,257 reverse their direction of risk between cancers. A signature sold as pan-cancer inherits that error rate.



Recurrent genes by direction · 17 replicated cohorts · adjusted P < 0.05

11,333 associations across 17 replicated cohorts

11,333

Gene–cohort associations

Each one a Cox model fitted per gene, at adjusted P < 0.05 within its cohort.

2,611

Recurrent genes

Significant in two or more cohorts, each with its direction recorded per cancer.

17

Replicated cohorts

13 transcriptome and 4 proteome cohorts clearing the replication rule.

2

Modalities, cross-checked

RNA and antibody-based protein, letting one confirm or contradict the other.

TCGA · 34 cohorts entering the screen · primary recurrence set

Every cohort is validated on held-out patients

Every cohort is split 80 / 20 before any model is fitted. A cohort enters the primary set only if it clears all three gates. Cohorts that fail are reported, not dropped quietly.

GATE ONE

10 genes

At least ten genes significant after correction within the cohort.

GATE TWO

0.70 agreement

Seven in ten significant genes keep their direction on held-out patients.

GATE THREE

10 events

At least ten events in the test split, so agreement is not an artefact of size.

Of the cohorts screened, 17 qualified. Three are flagged underpowered, six exploratory, and the rest carry minimal signal — each labelled rather than removed.

Tier rule applied uniformly across RNA and RPPA platforms

High proliferation predicts worse survival in every cancer

One coherent axis: the mitotic spindle and checkpoint, the G2/M driver FOXM1, the anti-apoptotic BIRC5, and nucleotide supply for replication.

Gene	Role	Cohorts	Median HR per SD
TPX2	Spindle assembly, activates Aurora A	9 of 9	1.64
UBE2C	APC/C ubiquitin ligase, mitotic exit	8 of 8	1.56
KIF20A	Mitotic kinesin, cytokinesis	8 of 8	1.56
AURKB	Chromosome passenger kinase	8 of 8	1.47
CDC20	APC/C activator, spindle checkpoint	7 of 7	1.61
CEP55	Cytokinesis, abscission	7 of 7	1.53
MKI67	Proliferation marker, Ki-67	7 of 7	1.47

Cohorts significant of cohorts replicating on held-out patients · hazard ratio per standard deviation of expression

Retained tissue identity predicts better survival

Gene	Cohorts	Median HR per SD
CBX7	5 of 5	0.59
ALDH2	4 of 4	0.63
ATP8A1	4 of 4	0.64
NDRG2	4 of 4	0.66
CLEC3B	5 of 4	0.66
ACSL5	4 of 4	0.68
HLF	5 of 5	0.70
ABAT	4 of 4	0.72

Polycomb and quiescence

CBX7 is the most recurrent protective gene in the atlas, protective in five cancers and replicating in all five.

Circadian clock output

HLF, TEF, DBP, PER2 and PER3 are protective wherever significant, with no risk-associated hit. The core activators carry no signal, so this is loss of rhythmic output rather than loss of the clock.

Retained metabolism

ALDH2, ABAT and FBP1 describe a gradient from differentiated oxidative metabolism to dedifferentiated, glycolytic tumours.

Cohorts significant of cohorts replicating · 426 genes protective in two or more replicated cohorts

The same gene can be protective in one cancer and harmful in another

For each gene, the number of cohorts where it raises risk against the number where it protects. A single pan-cancer sign is wrong in the smaller column every time.

Gene	Cohorts significant	Risk-associated in	Protective in
ANXA2	6	5	1
FLNC	6	5	1
GBP2	6	4	2
OAS1	6	4	2
SNCG	6	4	2
APOL1	6	3	3
CYP27A1	6	1	5

1,257 of 2,611 recurrent genes are discordant · directions replicate on held-out patients within each cohort

The immune microenvironment drives the reversals

Immune-responsive

Protective

Melanoma, head and neck, endometrial, bladder. High mutation burden or viral antigens, where infiltrating cytotoxic T cells and interferon signalling reflect active anti-tumour immunity.

Inflammatory or suppressive

Risk-associated

Glioma, clear-cell renal, prostate. The same genes mark the mesenchymal, macrophage-rich state; in renal cancer dense but exhausted CD8 infiltrate predicts worse outcome.

Hypoxia, reversed

Cancer-specific

In clear-cell renal cancer, VHL loss makes HIF activity constitutive, so high HIF-target expression marks the better-prognosis well-differentiated subtype.

Stroma, by contrast, does not reverse: extracellular matrix and EMT genes are risk-associated in 66 of 68 hits. The atlas separates what transfers from what must be set per indication.

Each reading is established tumour immunology; the atlas supplies the direction and the cohort

Gene expression improves prediction in 12 of 17 cohorts

12 of 17

Replicated cohorts where the gene model beats clinical covariates alone

+0.076
Mean C-index gain

+0.33
Largest single cohort

Across all tiers with sufficient test events the mean gain falls to about +0.02. The atlas earns its value by identifying which indications are tractable, not by claiming uniform lift.

Cohort	Clinical	Gene model	Gain
MESO	0.53	0.86	+0.33
SARC	0.55	0.79	+0.24
LAML	0.62	0.77	+0.15
PRAD	0.65	0.78	+0.13
LGG	0.78	0.88	+0.10
HNSC	0.56	0.64	+0.08
LUAD	0.64	0.62	−0.02

Held-out concordance index · replicated cohorts with at least ten test events · clinical covariates are stage, grade and demographics

Three candidates are ready for external validation

Candidate	Intended use	Evidence	Indications
Proliferation score	Pan-cancer risk marker	Every hit risk-associated; proteins agree at G2/M	MESO, LGG, LIHC, LUAD, PRAD, SKCM, KIRC
Circadian output score	Pan-cancer favourable marker	19 protective hits, none risk-associated	KIRC, KIRP, LGG, LIHC, LUAD, MESO, PRAD
IGFBP2	Glioma and renal risk marker	RNA and protein both risk-associated	LGG, GBM, KIRC
PAI-1	Invasion and stroma marker	RNA and protein both risk-associated	KIRC, GBM, LGG
Immune score	Context-dependent; sign set per cancer	Protective in four cancers, risk in three	Not usable as one pan-cancer marker
CBX7	Tumour-suppressor candidate	Protective in five cancers, all replicating	KIRC, LGG, MESO, PAAD, SKCM

The first three combine recurrence across cancers, held-out replication and independent protein support. None has been validated outside TCGA.

Tiered by intended use · protein support from antibody-based measurement in the same cohorts

No external validation yet; replication is the next step

OPEN RISKS

- No external validation. Every figure is internal to TCGA, split 80 / 20.
- The per-gene screen does not adjust for stage, grade or molecular subtype.
- Gene sets were curated by hand, not by formal enrichment testing.
- Two cohorts rest on fewer than ten events and are reported separately.
- Associations and directions are established; mechanism is not.

NEXT MILESTONE

Replicate in an independent cohort

The proliferation score, the circadian output score and IGFBP2 are the three to test first. Each has a stated direction, a stated indication set and held-out support inside TCGA, so a replication study has a pre-specified hypothesis rather than a fishing licence.

Two hypotheses follow directly and can be tested on the same data: whether a proliferation score and an immune score with a cancer-specific sign together explain most cross-cancer variation, and whether clock output stays protective after adjusting for proliferation.